Ahmad Bashir

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Address: Lahore, Pakistan (Home)

ABOUT ME

Machine Learning Engineer with over 3 years of experience in computer vision, deep learning, and real-time video processing. Skilled in developing and optimizing AI-driven solutions, including generative AI, 3D vision, and AI avatars, using Python, PyTorch, and OpenCV. Currently leading innovative projects at Colossyan as a Research Engineer, with a BS in Computer Science from ITU Lahore. Passionate about advancing AI technologies.

WORK EXPERIENCE

COLOSSYAN INC. – BUDAPEST, HUNGARY

MACHINE LEARNING ENGINEER – 12/2022 – CURRENT

- Developed and optimized video generation models using PyTorch and diffusion techniques, improving visual realism and processing efficiency by 25%.
- Implemented deep learning techniques for avatar realism, gesture synthesis, and speech synchronization, resulting in a 30% increase in user engagement.
- Improved the accuracy and speed of real-time video background matting by 20%, enhancing visual quality in live video applications.

TRUVIZ.AI – NEW JERSEY, UNITED STATES

COMPUTER VISION ENGINEER – 12/2024 – 06/2025

- Designed AI-driven proctoring models with Python and OpenCV, enhancing real-time face tracking, gaze estimation, and anomaly detection.
- Implemented realtime proctoring models with a decent inference speed of 40fps.
- Integrated deepfake detection and scaled systems to monitor 10,000+ candidates, boosting reliability and accuracy.

INTELLIGENT MACHINES LAB ITU – LAHORE, PAKISTAN

RESEARCH ASSISTANT – 06/2021 – 11/2022

- Developed PyTorch-based deep learning models for 3D reconstruction, camera pose estimation, and NeRF preprocessing, improving point cloud accuracy by 15% and achieving 90% accuracy in initial camera pose estimation.
- Contributed to computer vision and robotics research, co-authoring a CVPR paper submission.

EDUCATION AND TRAINING

09/2018 – 06/2022 Lahore, Pakistan

BS IN COMPUTER SCIENCE Information Technology University of the Punjab, Lahore

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s): **ENGLISH** | **GERMAN**

SKILLS

Machine Learning Frameworks

TensorFlow | PyTorch

Programming Languages

C++ | Python

Data Science Tools

Pandas | Matplotlib | Numpy

Computer Vision Libraries

OpenCV | PCL | Scikit-Image | Pillow | TorchVision

Development Tools

VS Code | Anaconda | Jupyter Notebook | Git | Linux | AWS

Domains

Computer Vision | Artificial Intelligence | Deep Learning | Machine Learning | 3D Vision | AI Avatars | Diffusion Models | Generative AI

RECOMMENDATIONS

References will be furnished upon request.

PROJECTS

Selected Projects

● AI-Powered Video Interview Proctoring System (2024 – Present): Built a real-time AI-driven video interview proctoring system incorporating deepfake detection, gaze tracking, and facial behavior analysis.
● Cancerous Blood Cell Detection Model (2024): Developed a YOLOv11-based deep learning model for real-time detection of cancerous blood cells, achieving 95% accuracy on microscopic datasets.
● Real-Time Video Background Matting (2023): Optimized an AI model at Colossyan for seamless real-time video background removal, improving visual quality and processing efficiency.
● Neural Radiance Fields Without COLMAP (2022): Pioneered a novel NeRF-based 3D reconstruction approach that eliminated Structure-from-Motion (SfM) dependencies, enabling reconstruction of texture-poor surfaces.

HONOURS AND AWARDS

Competitions, Bootcamps & Workshops

● IEEE Xtreme Programming Competition by Institute of Electrical and Electronics Engineers (IEEE)
● Ignite CyberSecurity Hackathon Pakistan (Speed Programming) by Ignite Pakistan
● Superior Technical Symposium STS (Programming Competition) by Superior University Lahore

HOBBIES AND INTERESTS

Technology, Languages & Sports

Language Learning: Exploring new languages to enhance communication and cultural understanding.
Cricket: Playing and following the sport for teamwork, strategy, and fitness.

MANAGEMENT AND LEADERSHIP SKILLS

Project Leadership & Strategy

Technical Leadership: Leading AI/ML projects, guiding research, and making key technical decisions.
Project Management: Overseeing development, ensuring deadlines, and optimizing workflows.
Innovation & Problem-Solving: Developing novel AI solutions, refining models, and improving system performance.
Team Collaboration & Mentorship: Assisting and mentoring students, guiding junior engineers, and working with cross-functional teams.

Leadership & Event Organization

Event Management Lead: Google Developers Student Club ITU(2021)
Membership Team Lead: IEEE ITU Student Branch(2020)

COMMUNICATION AND INTERPERSONAL SKILLS

Teaching Assistant-Parallel and Distributed Computing

Mentored students, led problem-solving sessions, and facilitated discussions on advanced computing concepts, enhancing course delivery and student engagement.